Istio Hybrid: Multi-Cluster and VMs in a Service Mesh

Workshop on Istio in sidecar mode with the classic Istio APIs (VirtualService, DestinationRule, ServiceEntry, Gateway): several Kubernetes clusters and virtual machines in one shared mesh, on an IPv6-only network.

All material in this repository is in English.

Agenda

1. Fundamentals: service mesh and Istio
   - What is a service mesh, and what is Istio? Sidecar and ambient
   - Why a service mesh?
   - Challenges and benefits
   - Istio architecture and components
   - Istio ingress gateway vs. Kubernetes Gateway API
2. Explore the lab
   - Look around the lab: nodes, pods, sidecar, `istioctl proxy-status`
3. Ingress gateway on IPv6
   - Expose Bookinfo through the ingress gateway over IPv6
4. Two clusters, one mesh
   - Check the link between two clusters: `istioctl remote-clusters`
   - Answer requests with pods from the other cluster
5. Virtual machines in the mesh
   - WorkloadGroup, WorkloadEntry and EndpointSlice: who is who
   - Dual stack on the VM: what to set (`DUAL_STACK` and `ISTIO_DUAL_STACK`)
   - Add a virtual machine to the mesh (IPv6 only)
   - Ports of istiod and of the sidecar on the VM
   - Reference: Dual stack on the VM: details and measurements